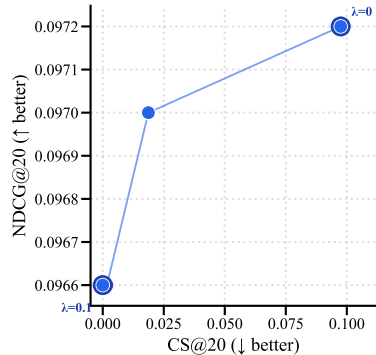


Pareto frontier:  $\lambda_c$  (inference-time penalty) trades off repetition (CS@20) for accuracy (NDCG@20)

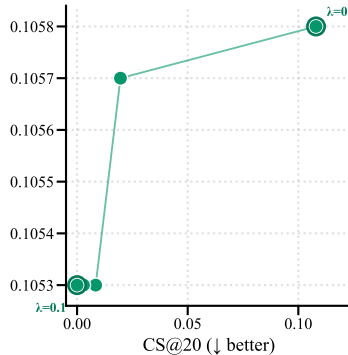


$\lambda_c$  increases  $\rightarrow$

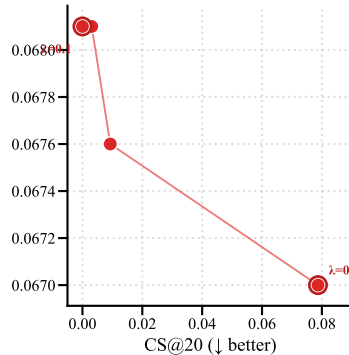
**TRIER**



**TRIER-C**



**TRIER-L**



**PACER-Full**

